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Dr. W.J. Cao

香 港 大 學
THE UNIVERSITY OF HONG KONG

CIVL2108 Principles of Civil Engineering Management

Dr. W.J. Cao
Department of Civil Engineering
Room 6-19A Haking Wong Building
wjcao@hku.hk

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Financial feasibility study -3

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1. Economic evaluation of facility investments

The basic concepts of facility investment evaluation, including time preference for consumption, opportunity cost, minimum attractive rate of return, cash flows over the planning horizon and profit measures.

Methods of economic evaluation, including the net present worth method, the equivalent annual value method, the benefit-cost ratio method, the internal rate of return method, and the incremental benefit-cost ratio method

Factors affecting cash flows, including depreciation and tax effects, price level changes (inflation and deflation), and treatment of risk and uncertainty.

Effects of different methods of financing on the selection of projects, including types of financing and risk, public policies on regulation and subsidies, the effects of project financial planning, and the interaction between operational and financial planning.

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2. Economic evaluation vs financing planning

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Economic evaluation is related to design, construction, operations and maintenance of the facility

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Financial planning requires knowledge of financial assets such as equities, bonds, notes and mortgages.

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3. Steps of economic evaluation

Generate	Establish	Estimate	Specify	Establish	Perform	Accept or reject
Generate a set of projects or purchases for investment consideration.	Establish the planning horizon for economic analysis.	Estimate the cash flow profile for each project.	Specify the minimum attractive rate of return (MARR).	Establish the criterion for accepting or rejecting a proposal, or for selecting the best among a group of mutually exclusive proposals, on the basis of the objective of the investment.	Perform sensitivity or uncertainty analysis.	Accept or reject a proposal on the basis of the established criterion.

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4. Risks and uncertainty

Business risks

- Market condition – uncertain about the period required to find an occupier
- Market changes – changes in conditions of supply and demand could affect the value
- Liquidity – unexpected need for cash is a risk factor as real property is difficult to dispose of on short notice

Financial risks

- Inflation – lower purchasing power & drive the cost up
- Interest rate movements – increase in interest rate will reduce the appraisal value of the development

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5. Sensitivity analysis

Determine the effect on the financial viability of the development as a result of the risks

Measure the impact on project outcomes of changing the key input values which are uncertain about

- Alter one variable each time to see the possible effect
- E.g. change the interest rate by ±1% each time
- Then, change other variables one by one in a similar manner to check for their sensitivity

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5. Sensitivity analysis

Private project

financial profitability of an investment

Public project

social benefits and social costs

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Social benefits and social costs

- There is an existing old and poor single-two road running between Town A and Town B. There is a proposal to construct a new highway to relieve the increasing traffic on the road. Identify the social benefits and social costs of the proposed highway project.

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6. Social benefits

- Less fuel used by vehicles due to shorter route taken between the two cities and reduced congestion on road
- Less travelling time taken due to shorter route, bigger road and less congestion
- Reduction of maintenance and repair costs to existing road--- materials and labour
- Fewer injuries to pedestrians
- Benefits as a result of greater turn-over in business because of better communication and transportation
- The prestige of society due to better infrastructure
- The opportunity for development in aspects such as education, recreation, entertainment, commerce and so on due to better transportation
- Reduction of noise, air and visual pollution to the area along the existing road.

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7. Social costs

- Cost of resources used in the highway construction
- Maintenance and repair costs to new highway
- Loss of crop production due to the fact that the land originally used for cultivation is now used as a highway
- Introduction of noise, air and visual pollution, social disruption and danger inflicted on residents around the new highway

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